Team 13 Project 2: UML Modeling

Header Table:

Name	CWID	Crimson Email	Hours Worked
Margaret Anderton (submitter)	12307232	mranderton@crimson.ua.edu	4
Wyatt Barras	12560255	wabarras@crimson.ua.edu	5
Connor Bellard	12478890	cjbellard@crimson.ua.edu	
Aidan Stanley	12404651	atstanley@crimson.ua.edu	3
Danni Sumner	12211512	dasumner@crimson.ua.edu	

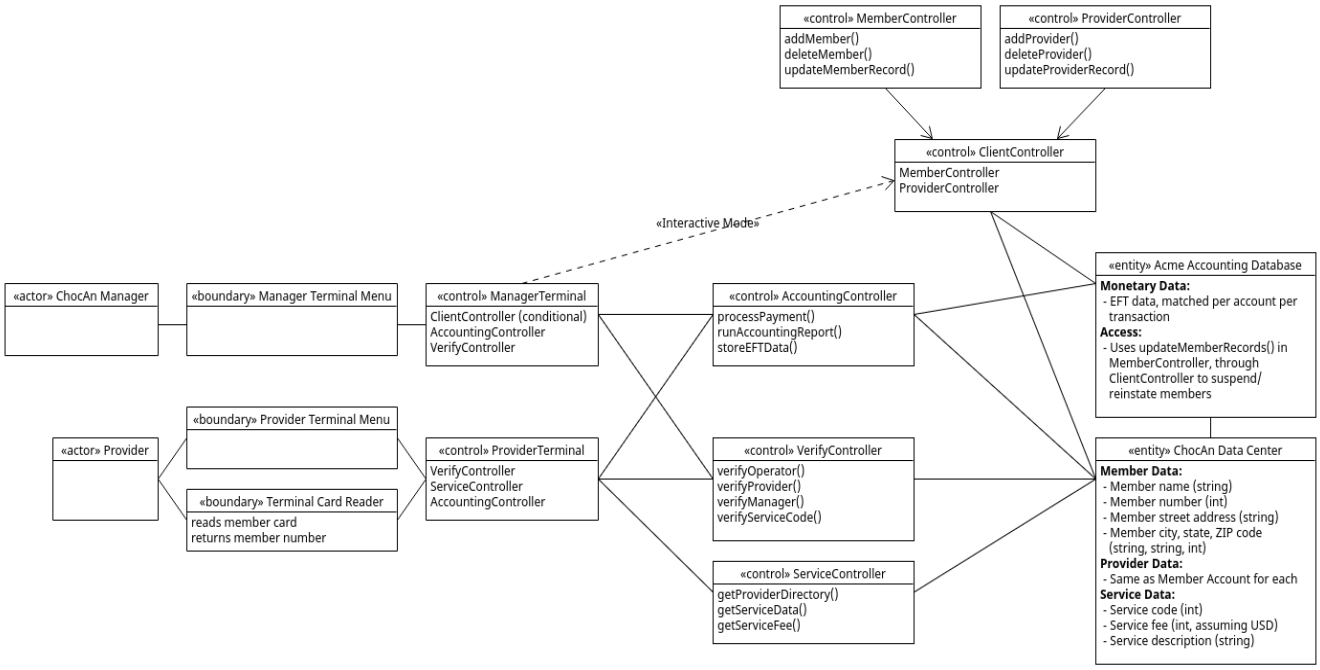
Task Distribution:

Name	Tasks Performed	Percentage
Margaret Anderton	Access Provider Directory, Bill ChocAn, Receive Healthcare activity diagrams	20%
Wyatt Barras	Stereotyped Class Diagram, putting document together	20%
Connor Bellard	Activity Diagram (overall), analysis paragraph	20%
Aidan Stanley	Verify Operator, Manager, Provider activity diagrams	20%
Danni Sumner		20%

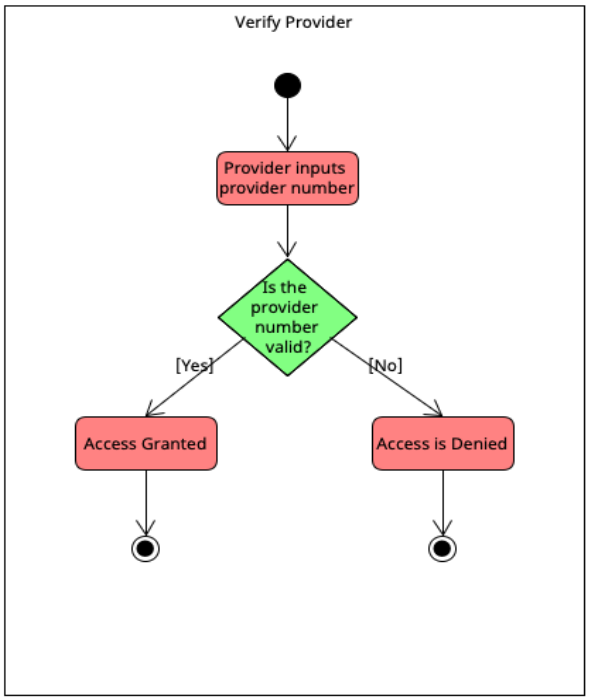
Analysis:

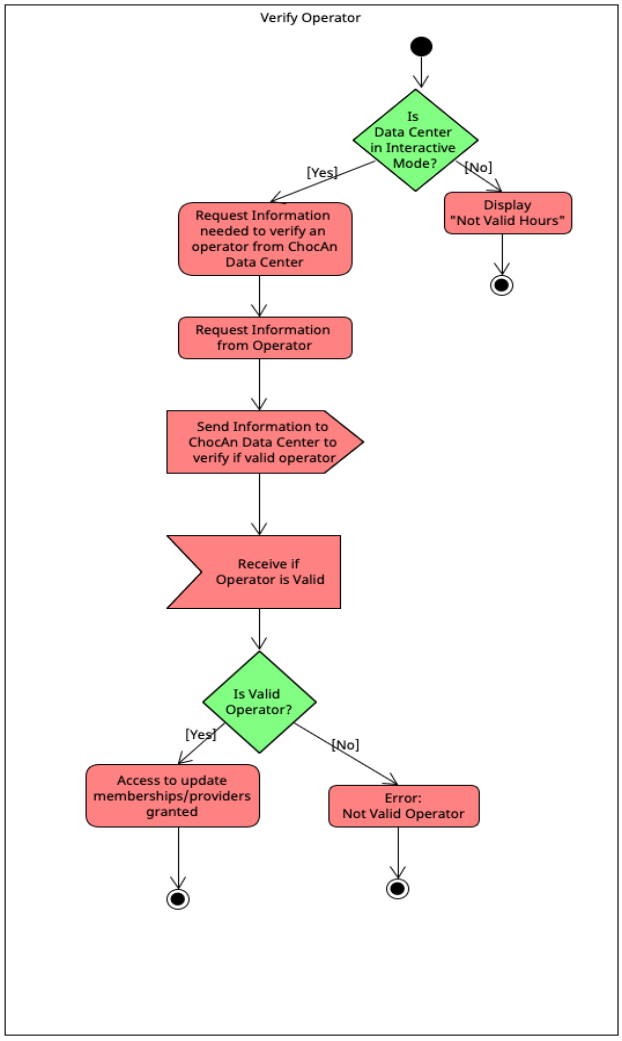
The ChocAn Processing System will serve as the primary means of interaction between customers and Chocoholics Anonymous. All mentioned use cases will need to be computerized due to the use of emails to send reports and the storage of provided services & membership records being digital. The data will be stored in servers at the physical ChocAn Center; should ChocAn possess multiple physical locations, the data will be stored on servers at a designated physical Data Center. To this end, physical server racks and SSD devices will be needed for the ChocAn Processing System to contain its information. Other hardware would be connected to the System (provider terminals, manager terminals, Internet-routing cables for these terminals and to connect to ACME, and membership card readers) and would be necessary for the interaction with these aforementioned systems. Users would then interact with these external systems in order to interact with the main ChocAn Processing System.

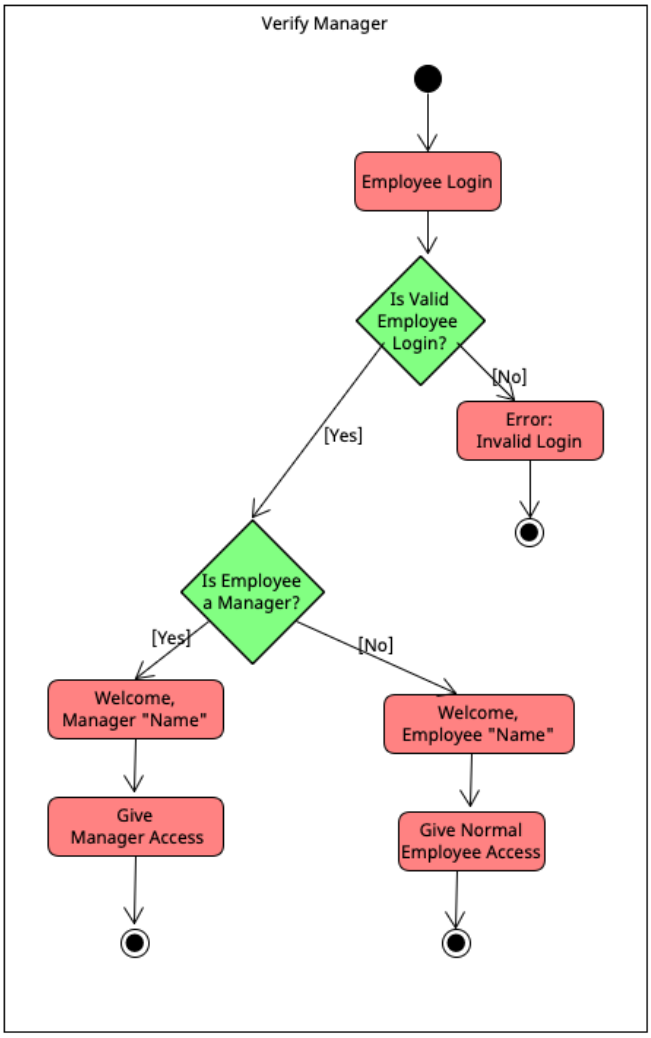
Stereotyped Class Diagram:

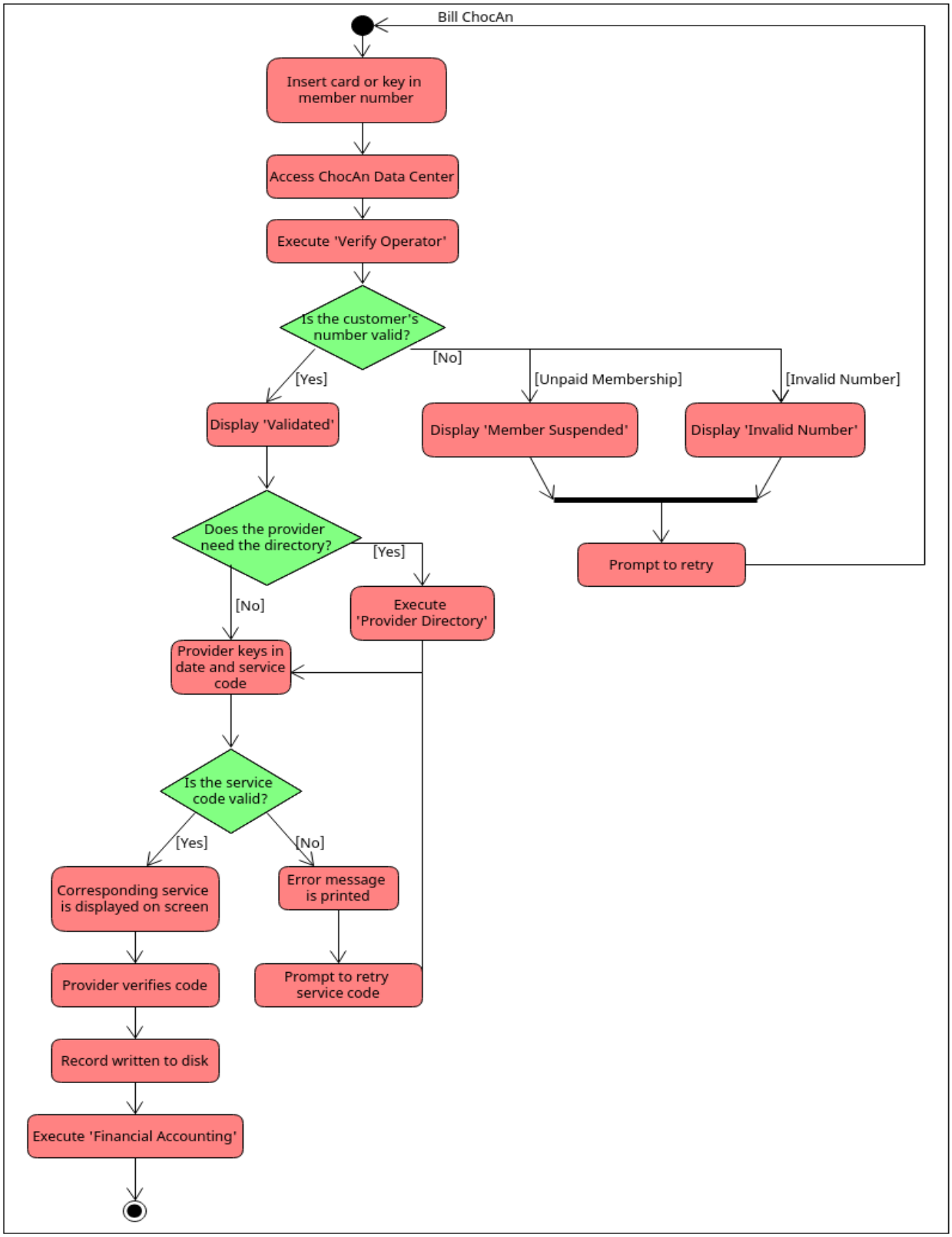


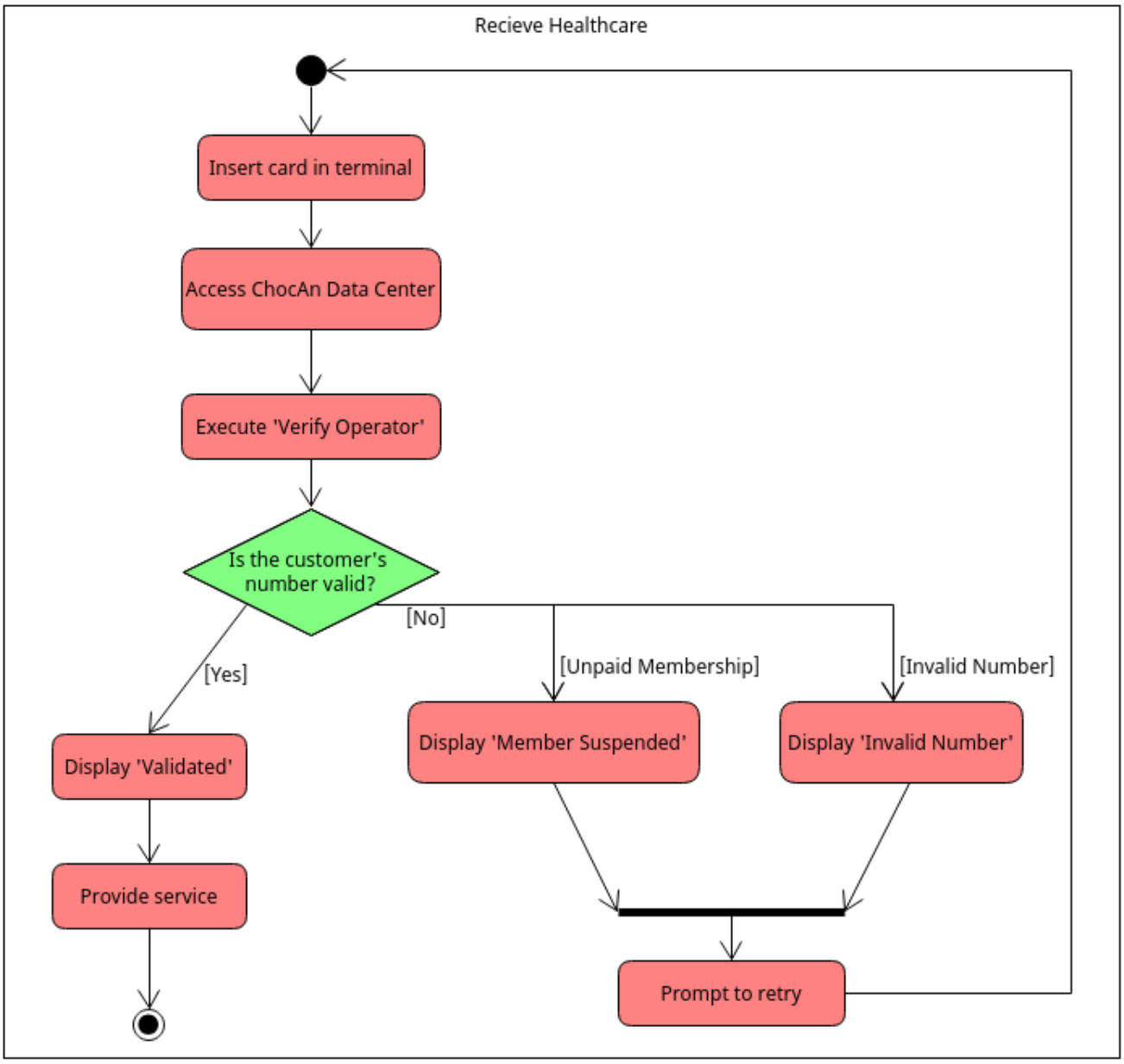
Activity Diagrams:

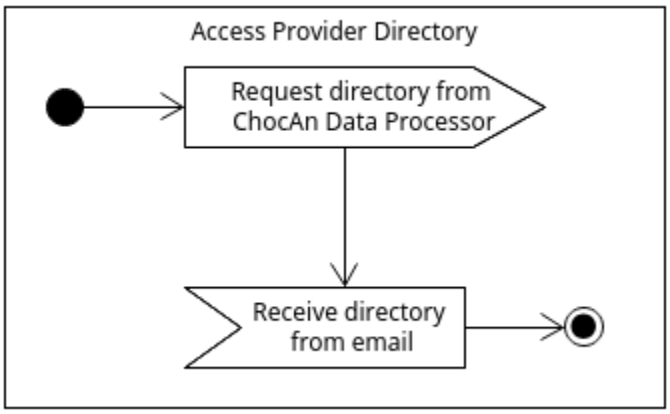


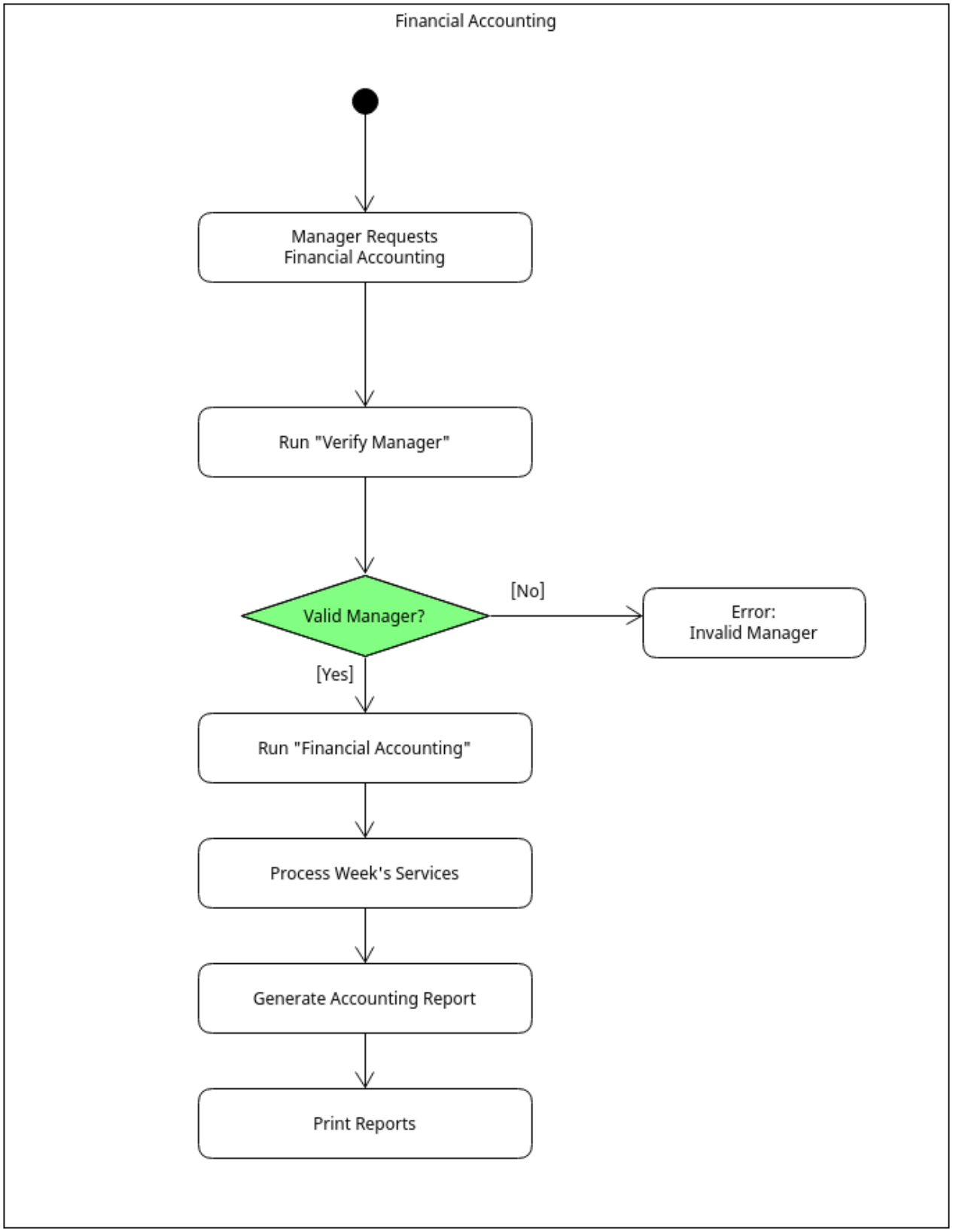

Verify Operator
Is Data Center in Interactive Mode?
[Yes]
[No]
Request Information needed to verify an operator from ChocAn Data Center
Display "Not Valid Hours"
Request Information from Operator
Send Information to ChocAn Data Center to verify if valid operator
Receive if Operator is Valid
Is Valid Operator?
[Yes]
[No]
Access to update memberships/providers granted
Error: Not Valid Operator

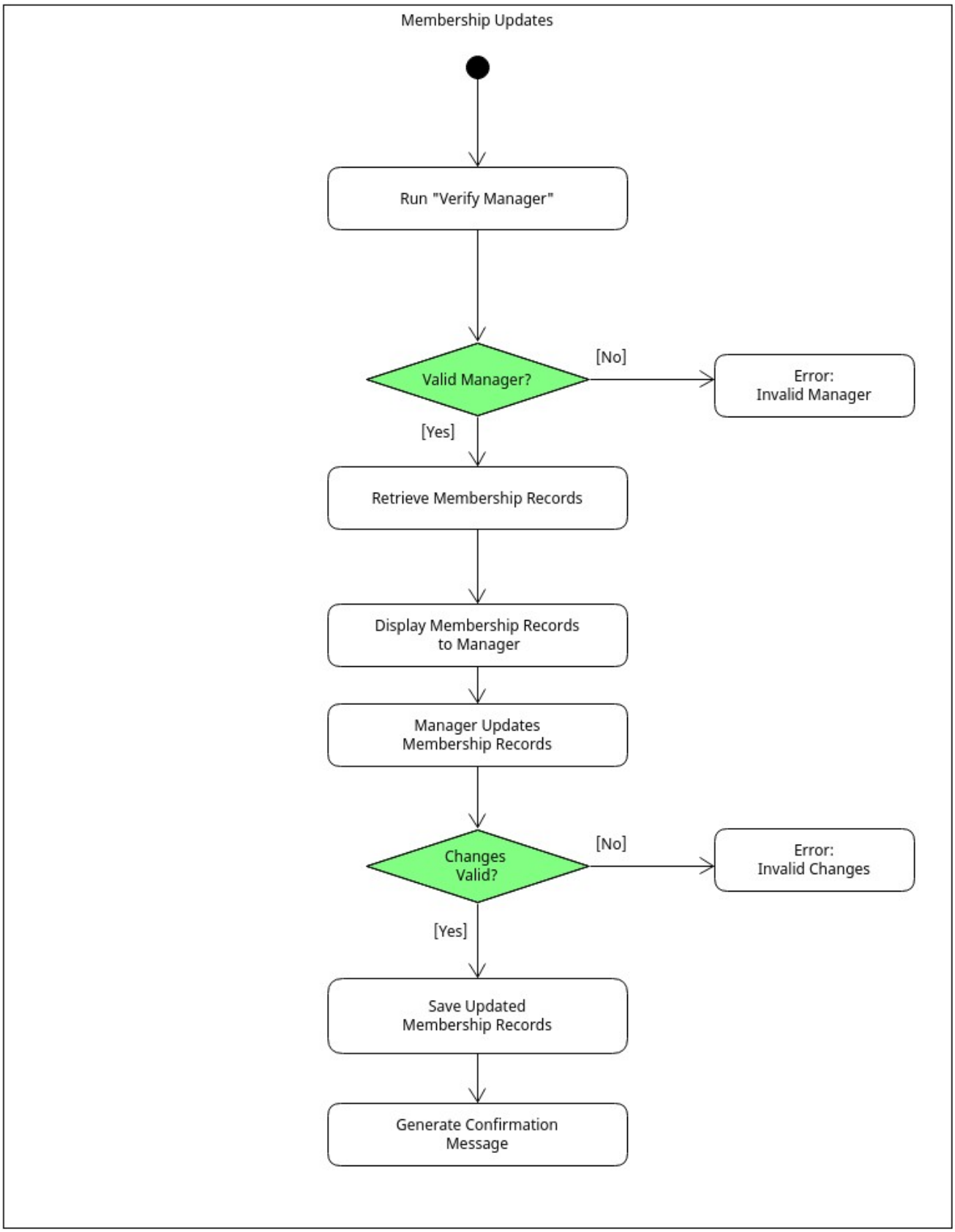

Verify Manager
Employee Login
Is Valid Employee Login?
[No]
[Yes]
Error: Invalid Login
Is Employee a Manager?
[Yes]
[No]
Welcome, Manager "Name"
Welcome, Employee "Name"
Give Manager Access
Give Normal Employee Access


Bill ChocAn
Insert card or key in member number
Access ChocAn Data Center
Execute 'Verify Operator'
Is the customer's number valid?
[Yes]
[No]
[Unpaid Membership]
[Invalid Number]
Display 'Validated'
Display 'Member Suspended'
Display 'Invalid Number'
Prompt to retry
Does the provider need the directory?
[Yes]
[No]
Execute 'Provider Directory'
Provider keys in date and service code
Is the service code valid?
[Yes]
[No]
Corresponding service is displayed on screen
Error message is printed
Provider verifies code
Prompt to retry service code
Record written to disk
Execute 'Financial Accounting'


Recieve Healthcare
Insert card in terminal
Access ChocAn Data Center
Execute 'Verify Operator'
Is the customer's number valid?
[No]
[Yes]
[Unpaid Membership]
[Invalid Number]
Display 'Validated'
Display 'Member Suspended'
Display 'Invalid Number'
Provide service
Prompt to retry


Access Provider Directory
Request directory from ChocAn Data Processor
Receive directory from email


Financial Accounting
Manager Requests
Financial Accounting
Run "Verify Manager"
Valid Manager?
[No]
Error:
Invalid Manager
[Yes]
Run "Financial Accounting"
Process Week's Services
Generate Accounting Report
Print Reports


Membership Updates
Run "Verify Manager"
Valid Manager?
[No]
Error:
Invalid Manager
[Yes]
Retrieve Membership Records
Display Membership Records
to Manager
Manager Updates
Membership Records
Changes
Valid?
[No]
Error:
Invalid Changes
[Yes]
Save Updated
Membership Records
Generate Confirmation
Message

Overall Activity Diagram:

